1: SIGNAL FLOW SUMMARY

1. The input RF signal is attenuated and protected, then amplified and filtered to reject unwanted signals.
2. The signal is mixed with the LO to down-convert to IF, then filtered and amplified.
3. The IF signal is detected (envelope/log detection) to convert to a voltage proportional to amplitude.
4. The video filter smooths the detected signal and sets the VBW.
5. The processed signal is displayed as a spectrum on the screen.
6. The control & processing section manages sweep, measurements, data processing, and user interface.

2: KEY PARAMETERS

- RBW (Resolution Bandwidth): IF bandwidth that determines frequency resolution
- VBW (Video Bandwidth): Smoothing bandwidth for the video signal
- Sweep Time: Time for one frequency sweep
- Span: Frequency range displayed
- Center Frequency: Center of the span
- Reference Level: Top of the display (dB)
- Attenuation: Input level attenuation

3: LEGEND (COLOR CODE)

- Front End (Attenuation & Protection)
- Tuning & Down Conversion
- Detection
- Video Filter
- Display
- Control & Processing